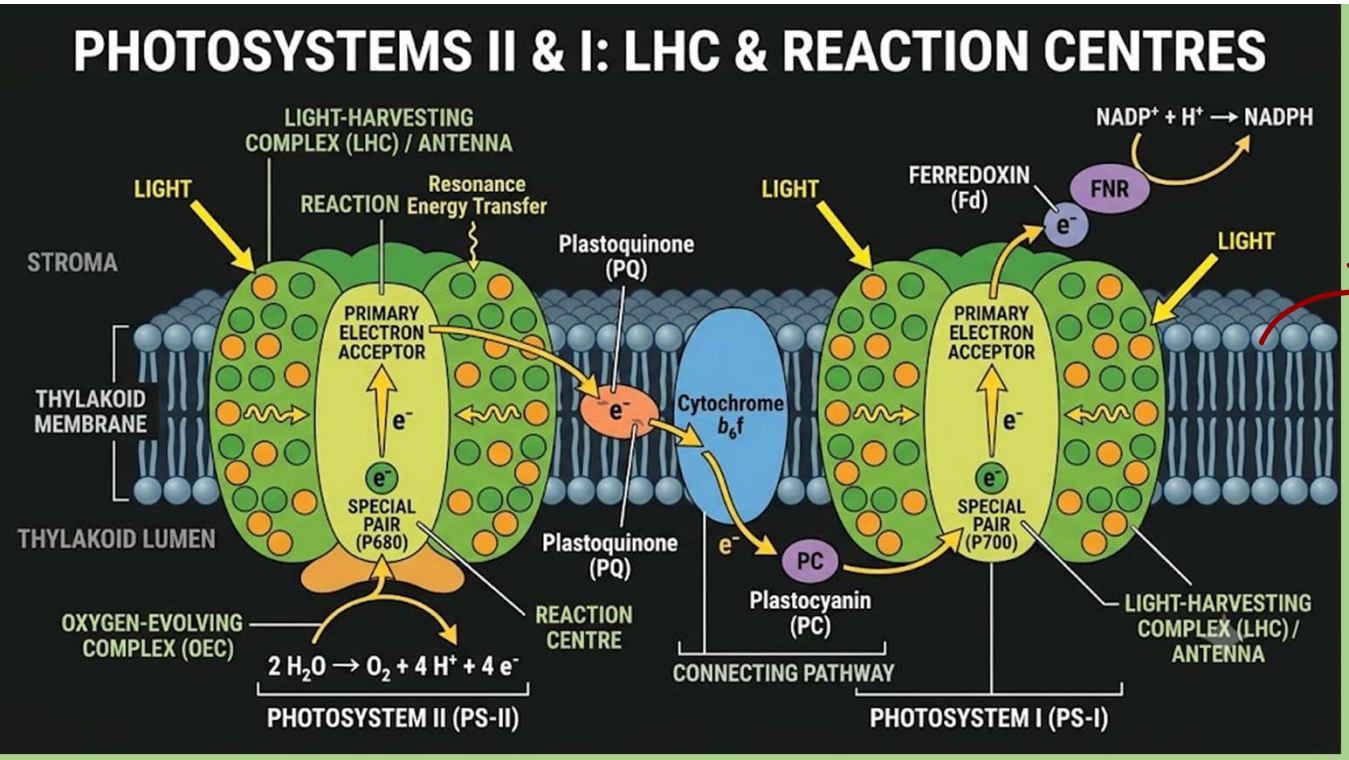
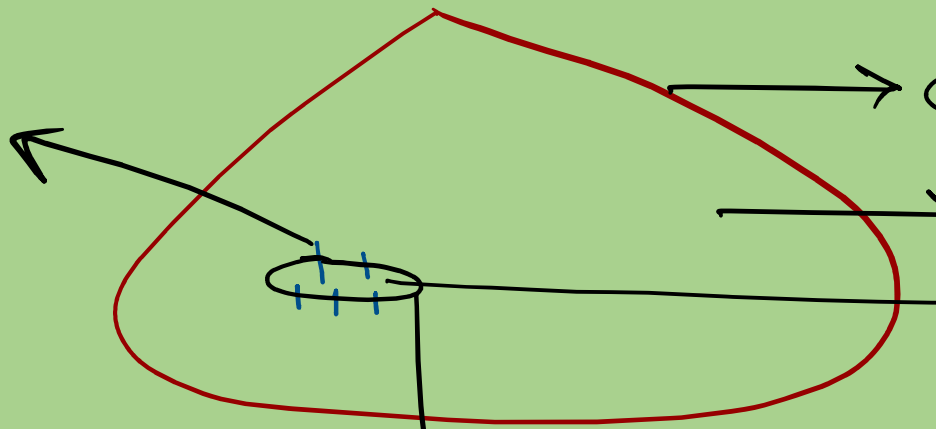


→ Stroma

→ Thylakoid membrane

→ Lumen

P.S.

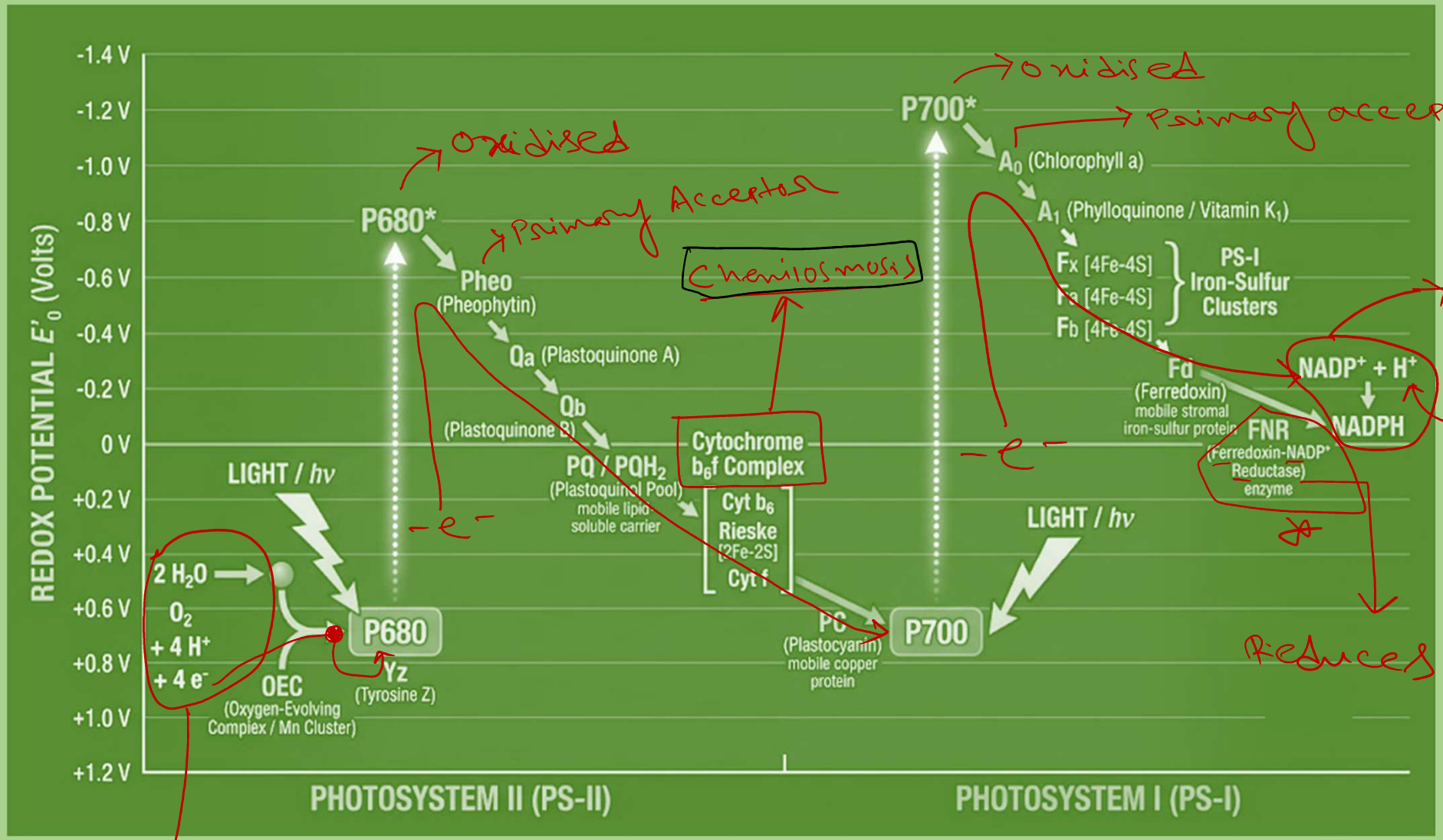


→ Chloroplast

→ Stroma

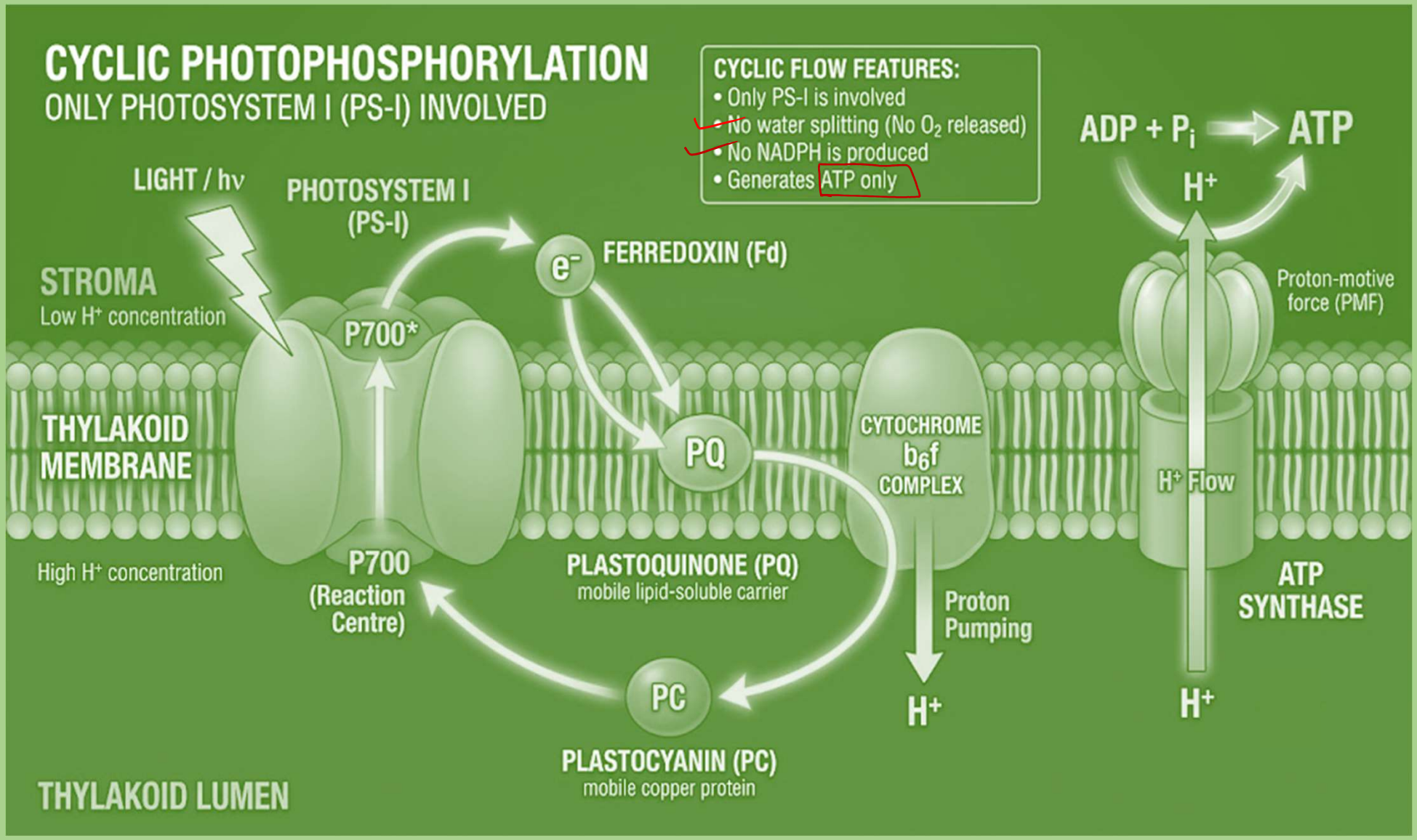
→ Lumen

→ Thylakoid membrane

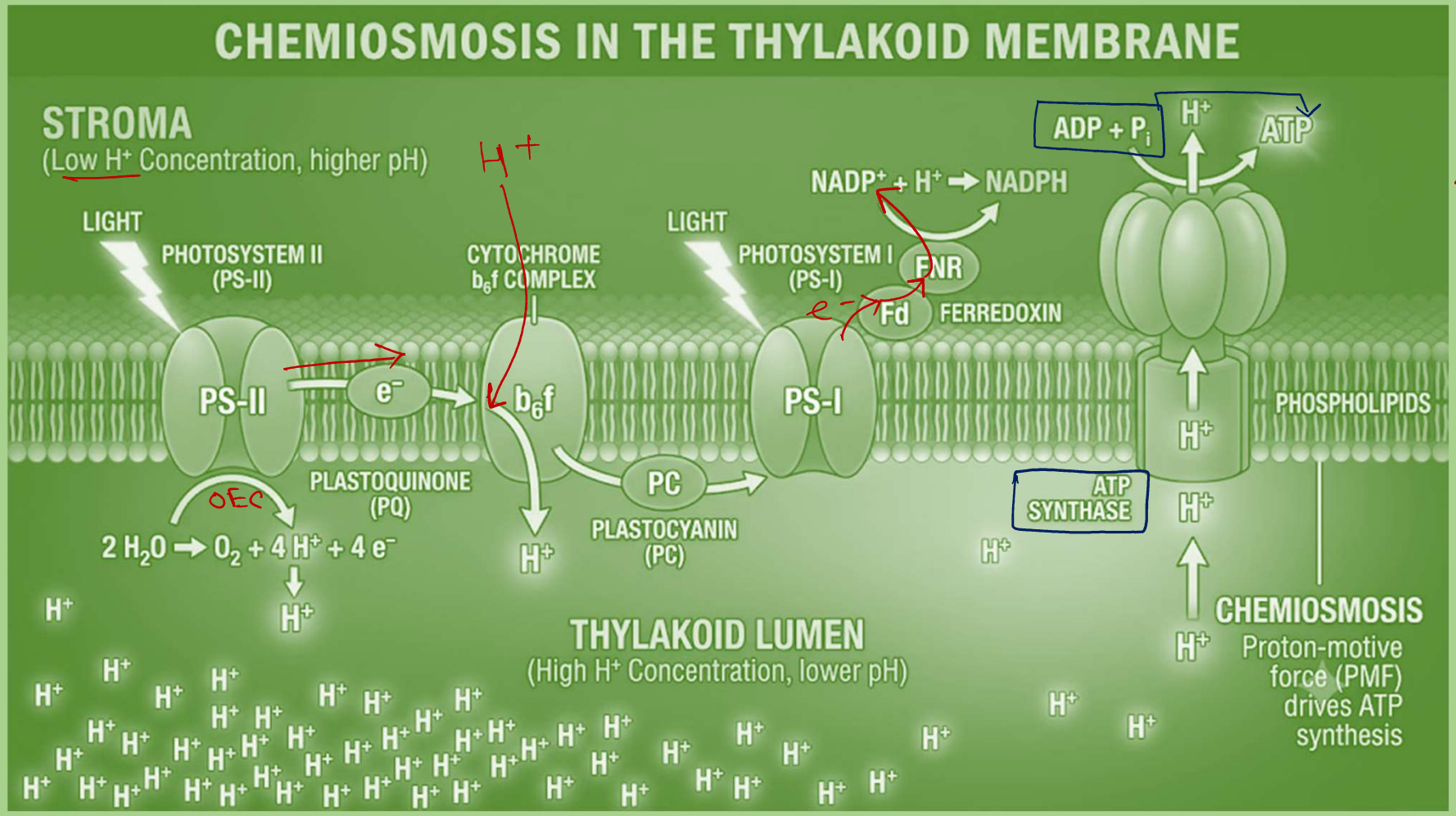


Thylakoid Lumen

Non-cyclic / Z-scheme



Cyclic - Photophosphorylation



Stroma

Proton

Motive

Force

Lumen

Chemiosmosis

Feature	Non-Cyclic Photophosphorylation (Z-Scheme)	Cyclic Photophosphorylation
Photosystems Involved	Both PS-II (P680) and PS-I (P700) work sequentially in series.	Only PS-I (P700) is involved; PS-II remains inactive.
Pathway of Electrons	Linear (unidirectional / Z-shaped): Electrons expelled from P680 never return to it.	Circular (closed loop): Electrons expelled from P700 return back to P700.
Ultimate Electron Source	Water (H_2O).	The P700 reaction center itself (the same electrons are continuously recycled).
Final Electron Acceptor	$NADP^+$(which gets reduced to NADPH).	Oxidized $P700^+$in the PS-I reaction center.
Complete Electron Route	$H_2O \rightarrow$ OEC $\rightarrow Y_Z \rightarrow$ P680 $\rightarrow$ Pheo $\rightarrow Q_A \rightarrow Q_B \rightarrow$ PQ $\rightarrow$ Cyt $b_6f \rightarrow$ PC $\rightarrow$ P700 $\rightarrow A_0 \rightarrow A_1 \rightarrow$ Fe–S $\rightarrow$ Fd $\rightarrow$ FNR $\rightarrow NADP^+$	P700 $\rightarrow A_0 \rightarrow A_1 \rightarrow$ Fe–S $(F_X, F_A, F_B) \rightarrow$ Fd $\rightarrow$ PQ $\rightarrow$ Cyt $b_6f \rightarrow$ PC $\rightarrow$ P700
Photolysis of Water	Occurs at the Oxygen-Evolving Complex (OEC) on the lumenal side of PS-II.	Does not occur.
Oxygen (O_2) Evolution	Evolved as a byproduct of water splitting.	Not evolved.
Net Products Formed	ATP, NADPH, and O_2.	ATP only (neither NADPHnor $O_2$is produced).
Site in Thylakoid Membrane	Occurs across the grana lamellae (appressed regions) and grana margins.	Occurs predominantly in the stroma lamellae (non-appressed regions), which lack PS-II and the FNR enzyme.
Proton (H^+) Gradient Source	Built by both water splitting in the lumen and H^+pumping via PQ/Cytochrome b_6f.	Built solely by H^+pumping via PQ/Cytochrome b_6f.
Wavelength of Light Required	Requires wavelengths both $\leq$ 680nm and 700nm to excite both photosystems.	Operates even under far-red light alone ($>$ 680 nm, up to 700nm).
Sensitivity to DCMU (Diuron)	Inhibited by DCMU (which blocks electron transfer between $Q_A$and $Q_B$in PS-II).	Unaffected by DCMU.
Physiological Role	Primary pathway for generating both reducing power (NADPH) and energy (ATP) for the Calvin cycle.	Supplies the extra ATP needed to meet the 3 ATP : 2 NADPH ratio of the Calvin cycle, and operates when $NADP^+$is depleted.